FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
CONCEPT RECAPITULATION TEST - IV
PAPER-2

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**A**	24.	**D**	47.	**B**
2.	**A**	25.	**B**	48.	**B**
3.	**A**	26.	**C**	49.	**B**
4.	**BD**	27.	**ABCD**	50.	**AC**
5.	**BC**	28.	**ABCD**	51.	**ABC**
6.	**ACD**	29.	**BCD**	52.	**ABC**
7.	**ABD**	30.	**ABCD**	53.	**ABCD**
8.	**BC**	31.	**AD**	54.	**BCD**
9.	**A**	32.	**B**	55.	**B**
10.	**D**	33.	**C**	56.	**C**
11.	**3**	34.	**9**	57.	**1**
12.	**3**	35.	**2**	58.	**1**
13.	**3**	36.	**7**	59.	**3**
14.	**1**	37.	**4**	60.	**8**
15.	**7**	38.	**7**	61.	**1**
16.	**2**	39.	**7**	62.	**7**
17.	**4**	40.	**6**	63.	**3**
18.	**1**	41.	**5**	64.	**5**
19.	**2**	42.	**8**	65.	**5**
20.	**9**	43.	**7**	66.	**9**
21.	**00002.99**	44.	**00145.80**	67.	**02011.25**
22.	**00001.66**	45.	**00002.00**	68.	**05142.85**
23.	**00000.25**	46.	**00070.31**	69.	**00833.33**

Physics

PART – I

SECTION – A

1. A

Rate of loss of water $= \int_0^2 \pi r^2 \left(dy\, \eta \sqrt{2gy} \right)$

2. A
3. A

$$\frac{2\lambda k}{r} = \frac{mv^2}{r}, T = \frac{2\pi r}{v}$$

4. BD
5. BC

For the bubble, PV=const.

$$\therefore \frac{\mathrm{dV}}{V} = \frac{-dP}{P}$$

$$\frac{dV}{V} = \frac{3dr}{r}, dP \approx -\frac{4T}{R},\ P \approx P_0 \left(\text{according to the question} \right)$$

$$\therefore\ dr = \frac{4Tr}{3P_0 R}$$

6. ACD
7. ABD
8. BC
9. A
10. D

[9-10]

For the charge

$$\vec{F} = q\left(\vec{v} \times \vec{B} \right) - b\vec{v}$$

$$\therefore \frac{d\vec{v}}{dt} = \frac{q}{m}\left(\frac{d\vec{r}}{dt} \times \vec{B} \right) - \frac{b}{m}\left(\frac{d\vec{r}}{dt} \right)$$

$$\therefore \Delta\vec{v} = \frac{q}{m}\left(\Delta\vec{r} \times \vec{B} \right) - \frac{b\left(\Delta\vec{r} \right)}{m}$$

SECTION – C

11. 3
12. 3
13. 3
14. 1
15. 7
16. 2
17. 4
18. 1

Sol: The change in internal energy of the gas is DU = 1.5 nR (DT)= 1.5 (1 mol) (253J/mol-K) (2K)
= 25 J.
The heat given to the gas = 42 J
The work done by the gas is DW = DQ – DU= 42 J – 25 J = 17 J.
If the distance moved by the piston is x, the work done is DW = (100 kPa) (8.5 cm2) x.
Thus, (105 N/m2) (8.5 × 10–4 m2) x = 17 J or, x = 0.2m = 20 cm.

19. 2
t1 = RC; t2 = R/L

$\rightarrow$LC = t1t2 = 0.1 sec -> T = 2pie $\sqrt{\frac{1}{LC}}$

T = 2 sec.

20. 9

SECTION – D

21. 00002.99
22. 00001.66
23. 00000.25

Chemistry

PART – II

SECTION – A

24. D
25. B
26. C
27 ABCD
28. ABCD
29. BCD
30. ABCD
31. AD
32 B
33. C

Solutions

$\underset{1}{A} \rightleftharpoons \underset{0}{B} + \underset{0}{C}$ $\qquad$ $\underset{\alpha}{C} \rightleftharpoons \underset{\alpha}{B} + \underset{0}{D}$

$1-\alpha \quad \alpha \quad \alpha$ $\qquad$ $\alpha-\beta \quad \alpha+\beta \quad \beta$

$1-\alpha \quad \alpha+\beta \quad \alpha-\beta$

$Kc_1 = \frac{(\alpha-\beta)(\alpha+\beta)}{1-\alpha}$ $\qquad$ $Kc_2 = \frac{\beta(\alpha+\beta)}{\alpha-\beta}$

$\frac{[C]}{[B]} = \frac{\alpha-\beta}{\alpha-\beta} = \frac{1}{5}$ $\qquad$ $\frac{\alpha}{\beta} = \frac{3}{2}$

Final Press=2[initial Press]

$\alpha+\beta=1$

So, $\alpha=\frac{3}{5}$ $\beta=\frac{2}{5}$

$Kc_1 = \frac{1}{2}$ $Kc_2=2$

$\frac{Kc_2}{Kc_1} = 4$

$\wedge_m = \frac{1000 \times 0.006}{0.5} = 12$

$\alpha = \frac{12}{200} = 0.06$

$C_1\alpha_1^2 = C_2\alpha_2^2$

$\alpha_2 = 0.135$

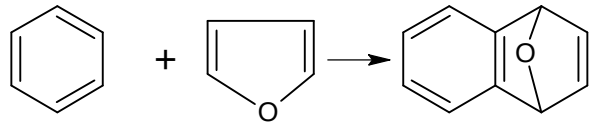
$i = 1.135$

$\pi = 12.72 atm$

SECTION – C

34. 9
Sol. check structure of glycine amino acid
35. 2
Sol. due to – I effect

36. 7



(P)

Sol.

37. 4

Sol. 4+6+6+12+12=40

$\therefore z = 4$

38. 7

Sol. $E^0_{Cl^-/AgCl/Ag} = E^0_{Ag^+/Ag} + \frac{0.0591}{1}\log K_{sp}$

$\therefore K_{sp} = 10^{-10}$ then let x be solubility of $AgCl$ in 0.01M solution of $NaCl$

$\Rightarrow 10^{-10} = x(x + 0.01)$

$x=10^{-8}$moles/L

39. 7

Sol. 4β and 6α particles are emitted.

40. 6

Sol. $NaCN, Na_3PO_4, Na_2CO_3, NaHCO_3, Na_2C_2O_4, Na_2HPO_4$ these are basic.

41. 5

Sol. m=3 and n=2

42. 8

Sol. Process AC=polytropic process P=KV

So molar heat capacity $C_m = C_v + \frac{R}{2} = 2R$

Process AB is isobaric $C_m = C_p = \frac{5R}{2}$

$$\frac{q_{AC}}{q_{AB}} = \frac{\int_{T_A}^{T_C} nC_m dT}{\int_{T_A}^{T_B} nC_{p,m} dT} = 0.8$$

43. 7

except Na_2CO_3 and $BaCO_3$ all other are correct

SECTION – D

44. 00145.80

Sol: $P^0_{liquid} = 748 - 648 = 100$ then $\frac{W_A}{W_B} = \frac{P^0_A M_A}{P^0_B M_B}$

$\therefore M_A = 145.8$

45. 00002.00

Sol. $Z = 1 + \frac{Pb}{RT}$ high pressure

So $\frac{dz}{dp}=\frac{b}{RT}=\frac{1}{2.8}\Rightarrow b=\frac{RT}{2.8}=\frac{22.4}{2.8}=4\left(N_A\times\frac{4}{3}\pi r^3\right)$

So $4\left(N_A\times\frac{4}{3}\pi r^3\right)$ for one mole of gas =2

46. 00070.31

Sol. m-eq. of $Ba(MnO_4)_2$ =m-eq.of H_2O_2

$\left(\because M=\frac{33.6}{11.2}\Rightarrow 3\right)$

$\frac{w}{375}\times 10\times 1000=3\times 125\times 2;$

w=28.125

% purity= $\frac{w}{40}\times 100$

$=\frac{28.125}{40}\times 100=70.31$

Mathematics

PART – III

SECTION – A

47. B

$$\int \underbrace{\frac{x^2}{(x\cos x-\sin x)}}_{f}\underbrace{\frac{dx}{(x\sin x+\cos x)}}_{g}$$

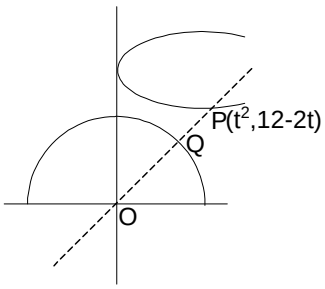
$$\frac{fg'-gf'}{fg}=\frac{g'}{g}-\frac{f'}{f}$$

$$=\ln\frac{g}{f}+c$$

48. B

$A\left(x_1,\sqrt{1-x_1^2}\right)$

$B\left(x_2,12-2\sqrt{x_2}\right)$



$$y=12-2\sqrt{x}$$

$$P\to(4,8)$$

$$\therefore OP=\sqrt{16+64}=\sqrt{80}$$

$$PQ=OP-OQ$$

$$=\sqrt{80}-1$$

49. B

50. AC

51. ABC

$A\to$ obvious

$B\to(\text{adj}A)^{-1}$

$$=\frac{adj(adjA)}{|adjA|}$$

$$\frac{|A|^{n-2}A}{|A|^{n-1}}=\frac{A}{|A|}$$

$C\to A^{-1}=A^T$

$\frac{1}{A}(cofactor\ matrix)^T = A^T$

$\rightarrow again\ transpose$

$\frac{1}{|A|}(cofactor\ matrix) = A$

If, $|A| = 1 \Rightarrow aij = Aij$

$and\ |A| = -1 \Rightarrow aij = Aij$

$D \rightarrow AA^T = A^T A = I$

$BB^T = B^T B = I$

$LHS\ \underbrace{(AB)B^T A^T}_{I}$

$\therefore AIA^T = AA^T = I$

$Similarly\ (A^{-1})(A^{-1})^T = A^{-1}(A^T)^T = A^{-1}A = I$

52. ABC

Sol. $\left(1-\frac{1}{2}\right)\left(1-\frac{1}{4}\right)\left(1-\frac{1}{6}\right)...\left(1-\frac{1}{2012}\right)$

$= \frac{1}{2}\times\frac{3}{4}\times\frac{5}{6}\times\frac{7}{8}\times.....\times\frac{2011}{2012} = \frac{2012!}{2^{2012}(1006!)^2}$

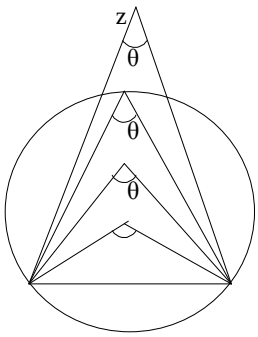
53. ABCD

Sol. point p on line L_1 $P(2+\lambda, 1+7\lambda, -2-5\lambda)$

Point P on line L_2 $P(4+r, -3+r, -r)$

$\Rightarrow \lambda = -1, r = -3$

54. BCD



(A) If $\arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2}$ then z_1 and z_2

Subtend right angle at circum centre origin.

$\therefore$ the chord joining z_1 and z_2 will subtend an angle θ at z such that

$\theta = \frac{\pi}{4}$ if $|z| = 1$

$\theta < \frac{\pi}{4}$ if $|z| > 1$, $\theta > \frac{\pi}{4}$ if $|z| < 1$

(B) $\left|z_1z_2+z_2z_3+z_3z_1\right|=\left|z_1\right|\left|z_2\right|\left|z_3\right|\left|\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}\right|$

$=\overline{\left|z_1+z_2+z_3\right|}$

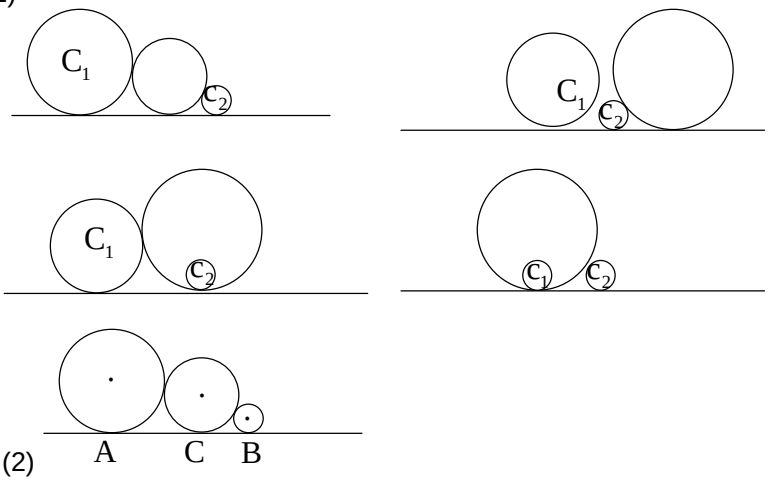
(C) $\left(\frac{(z_1+z_2)(z_2+z_3)(z_3+z_1)}{z_1z_2z_3}\right)\overline{\left(\frac{(z_1+z_2)(z_2+z_3)(z_3+z_1)}{z_1z_2z_3}\right)}$

(D)The triangle formed by joining z_1, z_3 and z_2 is isosceles and right angled at z_3.

55. B

56. C

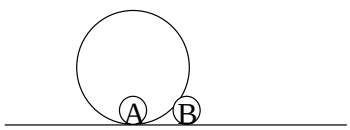
Sol. (1)



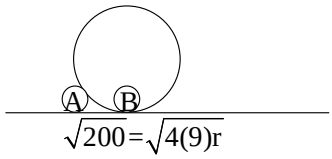
(2)

AB=AC+BC

$=\sqrt{225-25}=\sqrt{4r(9)}+\sqrt{4(r)(4)}$



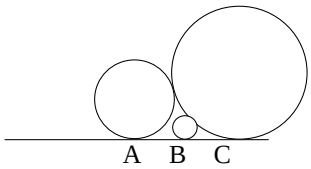
AB=AB



$\sqrt{200}=\sqrt{4(9)r}$

$\sqrt{4r(4)}=\sqrt{200}, r=\frac{50}{9}$

$r=\frac{25}{2}$



AC-BC=AB

$\sqrt{4(9)r}-\sqrt{4(4)r}=\sqrt{200}$

$2\sqrt{r}=\sqrt{200}$

r=50

SECTION – C

57. 1

58. 1

Sol: $T_n = \frac{H_{n+1} - H_n}{H_n H_{n+1}}$

As $H_{n+1} - H_n = \frac{1}{n+1}$

$\Rightarrow T_n = \frac{1}{H_n} - \frac{1}{H_{n+1}} \quad \Rightarrow T_1 = \frac{1}{H_1} - \frac{1}{H_2}$

$T_2 = \frac{1}{H_2} - \frac{1}{H_3}$

$\vdots \quad \vdots \quad \vdots$

$T_n = \frac{1}{H_n} - \frac{1}{H_{n+1}}$

$T_1 + T_2 + ... + T_n = \frac{1}{H_1} - \frac{1}{H_{n+1}}$

$\text{As } n \to \infty, \quad T_1 + T_2 + ... + 0 = 1$

59. 3

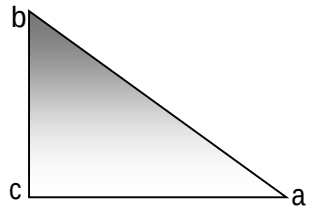
Sol. Given polynomial is a cubic with real coefficients $\Rightarrow$ at least one real root and other are conjugate complex zeroes.

Let $a = x + iy,\ b = x - iy,\ c = -2x$

$\because \quad a + b + c = 0$

$\therefore \quad |a^2| + |b^2| + |c^2| = 250$

$\Rightarrow \quad 3x^2 + y^2 = 125$



Using Pythagoras

$9x^2 + y^2 + 9x^2 + y^2 = 4y^2$

$\Rightarrow \quad 3x^2 = \frac{y^2}{3}$

$\therefore \quad \frac{4y^2}{3} = 125 \Rightarrow \frac{h^2}{125} = \frac{4y^2}{125} = 3$

60. 8

Sol. If a is the smallest element of such a set then 13-a is the largest and the remaining elements can be choosen from 12-2a elements i.e., 2^{12-2a}

Also, $13 - a \geq a \quad \Rightarrow a < 7$

$\therefore$ Number of such set $= 2^{10} + 2^8 + 2^6 + ... + 2^0 = 1365$

$= 3 \times 5 \times 7 \times 13$

$\therefore$ Divisors divisible by 5 is 8.

61. 1

Sol. Number of distinct terms in $\left(x + y + z + \frac{1}{xy} + \frac{1}{yz} + \frac{1}{zx} \right)^2$ is 18 and number of distinct terms in

$\left(x+y+z+\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)^2$ is 19

$\therefore \quad |m-n|=1$

62. 7

Sol. Clearly a team of 4 with two batsman and two bowler can be formed as

Case-I: 2 bowler+2batsman

Number of ways= ${}^4C_2\times{}^4C_2=36$

Case-II: 2 bowler+1 batsman +1 all-rounder

${}^4C_2\times{}^4C_1\times{}^1C_1=24$

Case-III: 1bowler +2 batsman +1 all-rounder

${}^4C_1\times{}^4C_2\times{}^1C_1=24$

$\therefore$ Probability that all-rounder is selected= $\frac{24+24}{36+24+24}=\frac{4}{7}$

$\therefore \quad p=7$

63. 3

Sol. For x-intercept y=0

$\therefore \quad \big||x-2|-a\big|=3$

$\Rightarrow \quad |x-2|-a=3 or -3$

$\Rightarrow \quad |x-2|=a+3 or a-3$

For 3 x-intercepts

a+3>0 and a-3=0 .....(i)

or a+3=0 and a-3>0 .....(ii)

From Eq. (i) a=3 and Eq. (ii) is rejected.

Hence, sum=3

64. 5

65. 5

66. 9

Sol. $\int_0^{27} xf'(x)dx=\lambda-3\int_0^3 x^2 f(x^3)dx$

$x^3=t$

$\Rightarrow 3x^2dx=dt$

$\int_0^{27} xf'(x)dx+\int_0^{27} f(x)dx=\lambda$

$\lambda=\int_0^{27}(f(x)+xf'(x))dx=(xf(x))_0^{27}$

$\lambda=27f(27)-0$

$=27\cdot\frac{1}{3}=9 \quad \{\because \text{f is decreasing}\therefore f(27)=\frac{1}{3}\}$

SECTION – D

67. 02011.25

Sol. dividend= x^{2007},

Divisor= x^2-5x+6

Let Q(x)be quotient, R(x)=ax+b be remainder

$x^{2007}=Q(x).(x^2-5x+6)+ax+b$

$x^{2007}=Q(x).(x-2)(x-3)+\underbrace{ax+b}_{R(x)}$ .....(i)

Now R(0)=b=?

Put x=2 in Eq. (i)

$2a+b=2^{2007}$ ....(ii)

Put x=3 in Eq. (i)

$3a+b=3^{2007}$ .....(iii)

Eq. (iii)-Eq.(ii) gives

$a=3^{2007}-2^{2007}$

Now $b=2^{2007}-2a=2^{2007}-2(3^{2007}-2^{2007})=2^{2007}+2.2^{2007}-2.3^{2007}$

$=2.3[2^{2006}-3^{2006}]=ab(a^c-b^c)$

Hence a=2;b=3;c=2006

68. 05142.85

Sol. Consonants →b,c,d,f,g,h (6)

Vowels →a,e,i,o,u (5)

W:

Case I: If word begins with consonants

then $({}^6C_4\times 4!)\times({}^5C_3\times 3!)=360\times 60=21600$

Case II:If word begins with vowels

$({}^5C_4\times 4!)\times({}^6C_3\times 3!)=120\times 120=12400$

Total=3600

69. 00833.33

Sol. $\because \int_a^x e^{f(t)}dt=\int_0^x g(x-t)dt+2x+3$

$\Rightarrow \int_a^x e^{f(t)}dt=\int_0^x g(t)dt+2x+3$ (Using King Property)

Differential both sides, we get

$e^{f(x)}=g(x)+2$

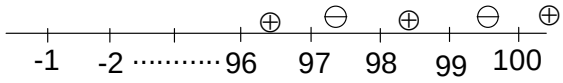
$\Rightarrow g(x)=e^{f(x)}-2$

$\Rightarrow g'(x)=e^{f(x)}.f'(x)$

$\because e^{f(x)}$ is always greater than zero.

$\therefore$ sign of g'(x) is same as sign of $f'(x)$.

$\therefore$ sign of g'(x)



Clearly, local extermum (maximum or minimum) will oxxur at x=99,97,95,….,3,1

$\therefore$ Sum of all the values=1+3+5+…+99=$\frac{50}{2}[2\times 1+(50-1)\times 2]=2500$